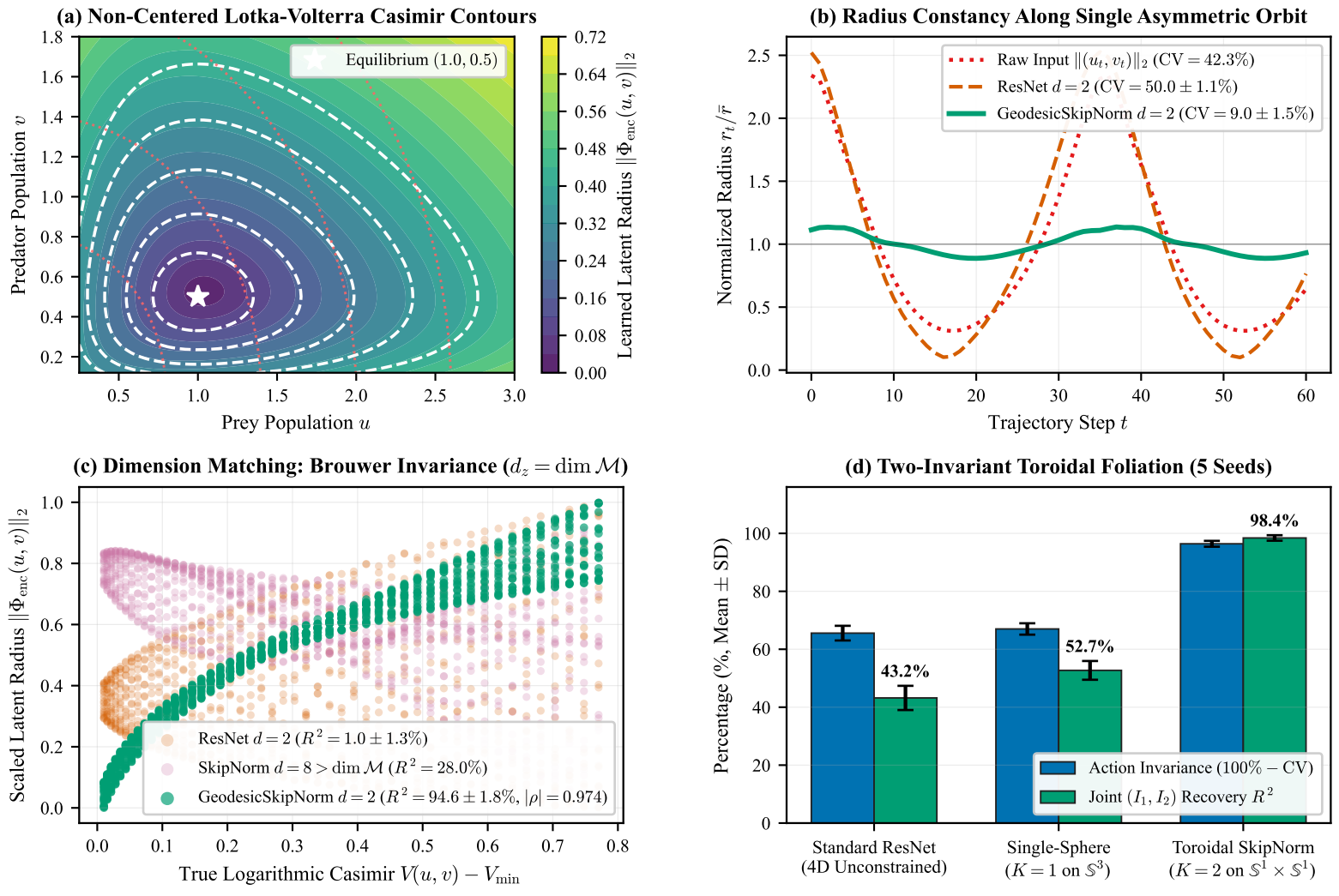

(a) Non-Centered Lotka-Volterra Casimir Contours
Equilibrium (1.0, 0.5)
Predator Population v
Prey Population u
Learned Latent Radius $\|\Phi_{\text{enc}}(u, v)\|_2$
(b) Radius Constancy Along Single Asymmetric Orbit
Raw Input $\|(u_t, v_t)\|_2$ (CV = 42.3%)
ResNet $d = 2$ (CV = 50.0 ± 1.1%)
GeodesicSkipNorm $d = 2$ (CV = 9.0 ± 1.5%)
Normalized Radius $r_t/\bar{r}$
Trajectory Step t
(c) Dimension Matching: Brouwer Invariance ($d_z = \dim \mathcal{M}$)
ResNet $d = 2$ ($R^2 = 1.0 \pm 1.3$%)
SkipNorm $d = 8 > \dim \mathcal{M}$ ($R^2 = 28.0$%)
GeodesicSkipNorm $d = 2$ ($R^2 = 94.6 \pm 1.8$%, $|\rho| = 0.974$)
Scaled Latent Radius $\|\Phi_{\text{enc}}(u, v)\|_2$
True Logarithmic Casimir $V(u, v) - V_{\min}$
(d) Two-Invariant Toroidal Foliation (5 Seeds)
43.2%
52.7%
98.4%
Action Invariance (100% − CV)
Joint (I_1, I_2) Recovery R^2
Percentage (%, Mean ± SD)
Standard ResNet
(4D Unconstrained)
Single-Sphere
($K = 1$ on $\mathbb{S}^3$)
Toroidal SkipNorm
($K = 2$ on $\mathbb{S}^1 \times \mathbb{S}^1$)